
MA 578: BAYESIAN STATISTICS

Homework 5

Due: November 18 at 12:30 PM (start of class)

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1. The file `glucose.dat` contains the plasma glucose concentration of 532 females from a study on diabetes.
 - (a) Make a histogram or kernel density estimate of the data. Describe how this empirical distribution deviates from the shape of a normal distribution.
 - (b) Consider the following mixture model for these data: For each study participant there is an unobserved group membership variable X_i which is equal to 1 or 2 with probability p and $1 - p$. If $X_i = 1$ then $Y_i \sim \mathcal{N}(\theta_1, \sigma_1^2)$, and if $X_i = 2$ then $Y_i \sim \mathcal{N}(\theta_2, \sigma_2^2)$. Let $p \sim \text{Beta}(a, b)$, $\theta_j \sim \mathcal{N}(\mu_0, \tau_0^2)$ and $\sigma_j^2 \sim \text{IG}(\nu_0/2, \nu_0\sigma_0^2/2)$ for both $j = 1$ and $j = 2$. Obtain the full conditional distributions of $(X_1, \dots, X_n), p, \theta_1, \theta_2, \sigma_1^2$ and σ_2^2 .
 - (c) Setting $a = b = 1$, $\mu_0 = 120$, $\tau_0^2 = 200$, $\sigma_0^2 = 1000$, and $\nu_0 = 10$, implement the Gibbs sampler for at least 10,000 iterations. Let $\theta_m^{(s)} = \min\{\theta_1^{(s)}, \theta_2^{(s)}\}$ and $\theta_M^{(s)} = \max\{\theta_1^{(s)}, \theta_2^{(s)}\}$. Using functions `acf()` and `effectiveSize()` from the package `coda` in R (or any other equivalent function in other languages), compute and plot the autocorrelation functions of $\theta_m^{(s)}$ and $\theta_M^{(s)}$, as well as their effective sample sizes.
 - (d) For each iteration s of the Gibbs sampler, sample a value $x \sim \text{Bernoulli}(p^{(s)})$, then sample $\tilde{Y}^{(s)} \sim \mathcal{N}(\theta_x^{(s)}, \sigma_x^{2(s)})$. Plot a histogram or kernel density estimate for the empirical distribution of $\tilde{Y}^{(1)}, \dots, \tilde{Y}^{(S)}$, and compare to the distribution in Part (a). Discuss the adequacy of this two-component mixture model for the glucose data.
2. In this problem, we will work with Stan on two simple models. We will use the `rstan` package. Make sure you have it installed and ready.
 - (a) Let our prior distribution be $p(\theta) = \text{Beta}(2, 2)$, and our sampling model be $p(y \mid \theta) = \text{Binom}(10, \theta)$. Suppose our observation is $y = 9$. We know from Chapter 3 that in this case, the posterior is $p(\theta \mid y) = \text{Beta}(11, 3)$. The goal is to make sure we know how to approximate this distribution using Stan.
 - i. Load the packages `rstan` and `bayesplot`.
 - ii. Define the Stan model as follows


```
data {
  real<lower=0> alpha;
  real<lower=0> beta;
  int<lower=1> n;
  int<lower=0, upper=n> Y;
}

parameters {
  real<lower=0, upper=1> theta;
}

model {
```

```

    Y ~ binomial(n, theta);
    theta ~ beta(alpha, beta);
  }

```

iii. Next, we will compile the model:

```

bb_sim <- stan(
  model,
  data = list(alpha = 2, beta = 2, Y = 9, n = 10),
  chains = 4, iter = 2000, warmup = 1000, thin = 1)

```

In this code, what do `chains`, `warmup`, and `iter` control?

iv. Using the following code, indicate (a) How many chains did we run? (b) What value did Stan use for burn-in? (c) Combining all chains, what's our total Markov chain sample size?

```

posterior_theta <- as.array(bb_sim, pars = "theta")
dim(posterior_theta)

```

Markov chain diagnostics:

- v. Using `mcmc_dens()` and `mcmc_hist()`, plot the posterior, and visually compare it to $\text{Beta}(11, 3)$
- vi. Using `mcmc_trace()`, construct trace plots of all 4 chains. Construct another set of trace plots that zooms in on the first 100 iterations. Summarize what you observe.
- vii. Using `mcmc_dens_overlay()` visually compare the density of the 4 chains.
- viii. Using `neff_ratio()` indicate the effective sample size ratio for our Markov chain sample and explain in words what this means.
- ix. Using `mcmc_acf()` describe in words the dependence among chain values and the mixing of the chain. What is the relationship between effective sample size and mixing time?

(b) Repeat Part (a) for prior distribution $p(\theta) = \text{Gamma}(3, 1)$, and sampling model $p(y \mid \theta) = \text{Poisson}(\theta)$, with data $(y_1, y_2) = (2, 8)$.

3. In the development of drugs and other chemical compounds, acute toxicity tests or bioassay experiments are commonly performed on animals. Such experiments proceed by administering various dose levels of the compound to batches of animals. The animals' responses are typically characterized by a dichotomous outcome: for example, alive or dead, tumor or no tumor. An experiment of this kind gives rise to data of the form

$$(x_i, n_i, y_i); \quad i = 1, \dots, k,$$

where x_i represents the i -th of k dose levels (often measured on a logarithmic scale) given to n_i animals, of which y_i subsequently respond with positive outcome. An example of real data from such an experiment is in the file `bioassay.csv`: twenty animals were tested, five at each of four dose levels.

We assume that the data points y_i are distributed as

$$p(y_i \mid \theta_i) = \text{Binom}(n_i, \theta_i),$$

where θ_i is the probability of death for animals given dose x_i , satisfying

$$\text{logit}(\theta_i) = \log \frac{\theta_i}{1 - \theta_i} = \alpha + \beta x_i$$

This is called logistic regression model. We also treat the outcomes in the four groups as independent of each other, given the parameters $(\theta_1, \theta_2, \theta_3, \theta_4)$.

Under this model, the sampling model will satisfy

$$p(y_i \mid \alpha, \beta, n_i, x_i) \propto \left[\frac{e^{\alpha + \beta x_i}}{1 + e^{\alpha + \beta x_i}} \right]^{y_i} \left[1 - \frac{e^{\alpha + \beta x_i}}{1 + e^{\alpha + \beta x_i}} \right]^{n_i - y_i}$$

Consider a non-informative prior $p(\alpha, \beta) \propto 1$. Therefore, the joint posterior distribution satisfies

$$p(\alpha, \beta \mid y, n, x) \propto \prod_{i=1}^k p(y_i \mid \alpha, \beta, n_i, x_i)$$

(We consider the sample sizes n_i and dose levels x_i as fixed for this analysis.)

- (a) Use random walk Metropolis algorithm to obtain posterior expectation and posterior standard deviation for the parameters α and β . (Run the simulations long enough to insure approximate convergence.)

Hint 1. It would be a good idea to use the output of a logistic regression, which is the maximum likelihood estimate of (α, β) in the above sampling model (likelihood function) for our four data points (that can be obtained using the following function in R) as the initial values for your chain:

```
gb <- glm(cbind(y, n - y) ~ x, family = binomial)
alpha <- coef(gb)[1]
beta <- coef(gb)[2]
```

Hint 2. The function `monitor()` from `rstan` package is a good friend.

- (b) Use the following functions from the `bayesplot` to evaluate the performance:

```
mcmc_trace()
mcmc_acf()
mcmc_dens_overlay()
mcmc_hist()
```

- (c) Repeat Parts (a) and (b) but use Stan instead to compare the performance.

Hint 3. Use the following Stan model code

```
data {
  int<lower=0> N;
  int<lower=0> n[N];
  int<lower=0> y[N];
  vector[N] x;
}
```

```
parameters {  
  real alpha;  
  real beta;  
}  
  
model {  
  y ~ binomial_logit(n, alpha + beta * x);  
}
```

Remark. Compute with log-densities: To avoid computational overflows and underflows, one should compute with the logarithms of posterior densities whenever possible. Exponentiation should be performed only when necessary and as late as possible; for example, in the Metropolis algorithm, the required ratio of two densities should be computed as the exponential of the difference of the log-densities.